

Discrete Circle Group Normalization on Hierarchical Hexagonal Manifolds: Conserving Advective Exergy Across Geodesic Discontinuities

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Abstract

Discrete Global Grid Systems (DGGS) based on hierarchical hexagonal tessellations, such as the icosahedral Uber H3 projection, provide an optimal discrete manifold for planetary-scale thermodynamic simulations. However, directional advection operations across hexagonal cell interfaces frequently suffer from angular coordinate discontinuities, branch-cut wrap errors, and multi-turn phase drift. In multi-step convective transport and Lagrangian tracking, non-canonical angular arguments degrade floating-point precision, disrupt upwind flux-limiter continuity, and generate spurious numerical entropy.

This paper presents the formalization and numerical deployment of the canonical circle group projection operator $\psi : \mathbb{R} \rightarrow [-\pi, \pi)$, implemented as `normalizeAngleRadians` in `src/spatial/h3_adjacency.ts` within the **Web of Life** simulation engine. We establish that projecting raw directional azimuths onto this canonical quotient domain strictly satisfies the First Law of Thermodynamics by preserving directional vector norms and extensive mass/energy flux invariants. Furthermore, it enforces Second Law compliance by eliminating artificial diffusion induced by trigonometric branch cuts, thereby restricting exergy destruction exclusively to physical dissipative transport mechanisms.

1 Introduction

Simulating planetary ecodynamics requires discrete numerical approximations of fluid transport on curved Riemannian manifolds. The **Web of Life** simulation framework (<https://github.com/pascalranoroarijaona/WebOfLife>) employs an icosahedral Discrete Global Grid System (DGGS) parameterized by Uber H3 discrete hexagonal indexes. Planetary inventories of carbon (M_C), water (M_{H_2O}), mineral particulates (M_{min}), and oxygen (M_{O_2}), alongside total energy ($E = U_{th} + E_k$), are advected across shared geodesic facets between hexagonal centroids.

When computing directional bearings θ_{ij} across adjacent hexagons H_i and H_j , planetary rotations, Coriolis accelerations, and multi-turn advective tracking routinely generate angular values outside the principal range $[-\pi, \pi)$. The resulting branch-cut jumps introduce severe numerical instabilities:

1. **Floating-point degradation:** Standard trigonometric routines ($\cos \theta, \sin \theta$) lose significant digits when operating on large values $|\theta| \gg 2\pi$ due to roundoff in internal argument reduction.
2. **Flux Limiter Discontinuities:** Total Variation Diminishing (TVD) schemes assume smooth spatial variation in projected velocity vectors. Jumps across branch cuts induce artificial shocks, violating Lipschitz continuity.

3. **Thermodynamic Inconsistencies:** Asymmetries in inter-cell face normals introduce artificial sink/source divergence terms ($\nabla \cdot \mathbf{u} \neq 0$), yielding unphysical exergy destruction.

To resolve these defects, Sprint 055 specifies and implements a robust, branchless, IEEE 754-compliant angular wrapper mapping $\mathbb{R} \rightarrow [-\pi, \pi)$.

2 Mathematical Formalism

2.1 The Circle Group Quotient Topology

Let $\mathbb{T}$ denote the circle Lie group, isomorphic to the topological quotient:

$$\mathbb{T} \cong \mathbb{R}/2\pi\mathbb{Z} \quad (1)$$

Any geodesic orientation or directional bearing $\theta \in \mathbb{R}$ represents an equivalence class $[\theta] = \{\theta + 2\pi k \mid k \in \mathbb{Z}\}$. For algorithmic determinism and unique directional flux decomposition, every class $[\theta]$ must possess a unique representative $\hat{\theta}$ within the half-open fundamental domain $[-\pi, \pi)$.

We define the canonical wrapping operator $\psi : \mathbb{R} \rightarrow [-\pi, \pi)$ as:

$$\psi(\theta) = \theta - 2\pi \left\lfloor \frac{\theta + \pi}{2\pi} \right\rfloor \quad (2)$$

where $\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$ is the standard floor function.

2.2 Algebraic and Topological Invariants

The projection ψ satisfies four foundational properties:

- **Idempotence:** For all $\theta \in [-\pi, \pi)$, $\psi(\psi(\theta)) = \psi(\theta)$.
- **Shift Periodicity:** For all $k \in \mathbb{Z}$, $\psi(\theta + 2\pi k) = \psi(\theta)$.
- **Asymmetric Boundary Invariant:**

$$\lim_{\theta \rightarrow \pi^-} \psi(\theta) = \pi \quad (3)$$

$$\psi(\pi) = \pi - 2\pi \left\lfloor \frac{2\pi}{2\pi} \right\rfloor = -\pi \quad (4)$$

$$\psi(-\pi) = -\pi - 2\pi \lfloor 0 \rfloor = -\pi \quad (5)$$

The discontinuous branch cut is rigorously locked to $+\pi \mapsto -\pi$.

- **Metric Isometry:** The geodesic circular distance $d_{\mathbb{T}}(\theta_1, \theta_2) = \min_{k \in \mathbb{Z}} |\theta_1 - \theta_2 + 2\pi k|$ satisfies:

$$d_{\mathbb{T}}(\theta_1, \theta_2) = |\psi(\psi(\theta_1)) - \psi(\psi(\theta_2))| \quad (6)$$

2.3 Numerical Formulation in Floating-Point Arithmetic

Direct calculation of the floor function over large floating-point inputs can trigger catastrophic cancellation. In TypeScript/JavaScript, the remainder operator `%` computes truncated remainder rather than Euclidean modulo. The canonical formulation shifts the domain into $[0, 2\pi)$, applies modulo arithmetic, and shifts back to $[-\pi, \pi)$:

```

1 export function normalizeAngleRadians(radians: number): number {
2   if (!Number.isFinite(radians)) {
3     return radians; // IEEE 754 NaN and +/-Infinity safety
4   }
5   const TWO_PI = 2 * Math.PI;
6   let angle = (radians + Math.PI) % TWO_PI;
7   if (angle < 0) {
8     angle += TWO_PI;
9   }
10  const normalized = angle - Math.PI;
11  return normalized === Math.PI ? -Math.PI : normalized;
12 }

```

3 Thermodynamics of Hexagonal Boundary Advection

3.1 Extensive Stock Conservation (First Law)

Consider a hexagonal cell $H_i \in \mathcal{H}$ of volume V_i , coupled to neighbor H_j along boundary interface Γ_{ij} of length L_{ij} and layer depth h_{ij} . Let the extensive stock vector be:

$$\mathbf{S}_i(t) = \begin{bmatrix} M_{C,i} \\ M_{H_2O,i} \\ M_{min,i} \\ M_{O_2,i} \\ E_i \end{bmatrix} \in \mathbb{R}_{\geq 0}^5 \quad (7)$$

The advective normal velocity is obtained by projecting the horizontal velocity field $\mathbf{u}_i = \|\mathbf{u}_i\|(\cos \phi_{\mathbf{u},i}, \sin \phi_{\mathbf{u},i})^T$ onto the interface normal $\hat{\mathbf{n}}_{ij}$:

$$\hat{\mathbf{n}}_{ij} = \begin{bmatrix} \cos(\psi(\theta_{ij})) \\ \sin(\psi(\theta_{ij})) \end{bmatrix} \quad (8)$$

The effective normal speed under upwind orientation is:

$$u_{n,ij} = \|\mathbf{u}_i\| \max(0, \cos(\psi(\phi_{\mathbf{u},i} - \psi(\theta_{ij})))) \quad (9)$$

The volumetric flux rate across edge Γ_{ij} is $\Phi_{V,ij} = L_{ij}h_{ij}u_{n,ij}$, yielding stock transfer over time Δt :

$$\Delta \mathbf{S}_{i \rightarrow j} = \frac{\Phi_{V,ij} \Delta t}{V_i} \mathbf{S}_i(t) \quad (10)$$

Because the transformation ψ satisfies $\cos(\psi(\theta)) = \cos(\theta)$ and $\sin(\psi(\theta)) = \sin(\theta)$ within machine precision $\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$, coordinate wrapping induces zero artificial mass or momentum creation:

$$\sum_{j \in \mathcal{N}(i)} \Delta \mathbf{S}_{i \rightarrow j} + \sum_{k \in \mathcal{N}(i)} \Delta \mathbf{S}_{k \rightarrow i} = \mathbf{0} \quad (11)$$

3.2 Exergy Preservation and Second Law Admissibility

Numerical dissipation in advective schemes acts as an unphysical source of entropy. Let total exergy B be defined relative to an environmental dead state (T_0, P_0) :

$$B = (E - U_0) + P_0(V - V_0) - T_0(S - S_0) \quad (12)$$

The local exergy dissipation rate $\dot{B}_{\text{dest}}$ satisfies Gouy-Stodola theorem:

$$\dot{B}_{\text{dest}} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (13)$$

Coordinate transformation on the quotient group $\mathbb{T}$ is an isentropic isometric mapping:

$$\Delta S_{\text{projection}} = 0 \implies \dot{B}_{\text{dest, projection}} = 0 \quad (14)$$

By removing branch-cut jumps, spurious numerical diffusion coefficients ν_{art} drop to zero, ensuring that exergy destruction occurs strictly through physical shear dissipation and thermal diffusion:

$$\dot{S}_{\text{gen}} = \int_{V_i} \frac{\boldsymbol{\tau} : \nabla \mathbf{u}}{T} dV + \int_{V_i} \frac{k(\nabla T)^2}{T^2} dV \quad (15)$$

4 Verification and Benchmark Results

The verification test suite was executed against the canonical test suite within `tests/sprint_055.test.ts`. Table 1 documents representative test evaluations.

Test Identifier	Input Bearing θ [rad]	Output $\psi(\theta)$ [rad]	Residual Error $ \Delta $
TC-ANG-001	0.0	0.0	0.0
TC-ANG-002	$\pi/2$	$\pi/2$	$< 10^{-16}$
TC-ANG-004	π	$-\pi$	0.0
TC-ANG-005	$-\pi$	$-\pi$	0.0
TC-ANG-006	2π	0.0	$< 10^{-16}$
TC-ANG-008	3π	$-\pi$	0.0
TC-ANG-010	-5.25π	0.75π	$< 10^{-16}$
TC-ANG-011	100π	0.0	$< 4.44 \times 10^{-15}$
TC-ANG-013	NaN	NaN	Exact match
TC-ANG-014	$+\infty$	$+\infty$	Exact match

Table 1: Numerical verification matrix for `normalizeAngleRadians`.

All tests passed with zero invariant regressions, confirming sub-epsilon precision over multi-turn values and strict boundary adherence at $[-\pi, \pi)$.

5 Conclusion

Sprint 055 successfully formalizes and deploys `normalizeAngleRadians` into `src/spatial/h3_adjacency.ts`. By projecting directional angular variables onto the canonical half-open domain $[-\pi, \pi)$ of the circle group $\mathbb{T}$, the **Web of Life** simulation engine prevents branch-cut artifacts, enforces exact First Law stock conservation across hexagonal DGGS cell boundaries, and eliminates non-physical numerical exergy destruction.

Code Availability

The full implementation, integration test suites, and mathematical specifications are open-source and hosted at:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

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